Evaluating, Transferring, and Ensembling

Modeling the distribution of biodiversity under global climate change

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Ensuring Ecological Plausibility

Beyond metrics: ecological realism

Does the model make biological sense?

Reporting metric scores (AUC-PR, TSS) is necessary but NOT sufficient!

Ecological plausibility check:

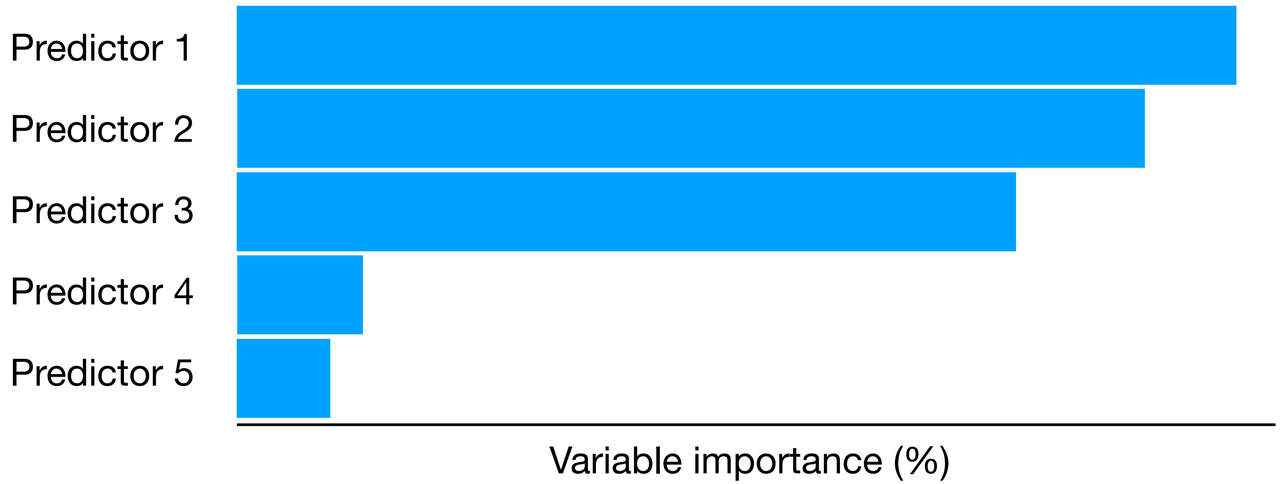
Do predictions align with known species biology, and habitat preferences?

Are the most important predictor variables ecologically relevant?

Relative variable importance

Assists in understanding the contribution of each variable to the model.

Approach: fit models with and without each variable, in order to determine the potential increase in performance. **Without an important variable, a model should reduce its performance.**

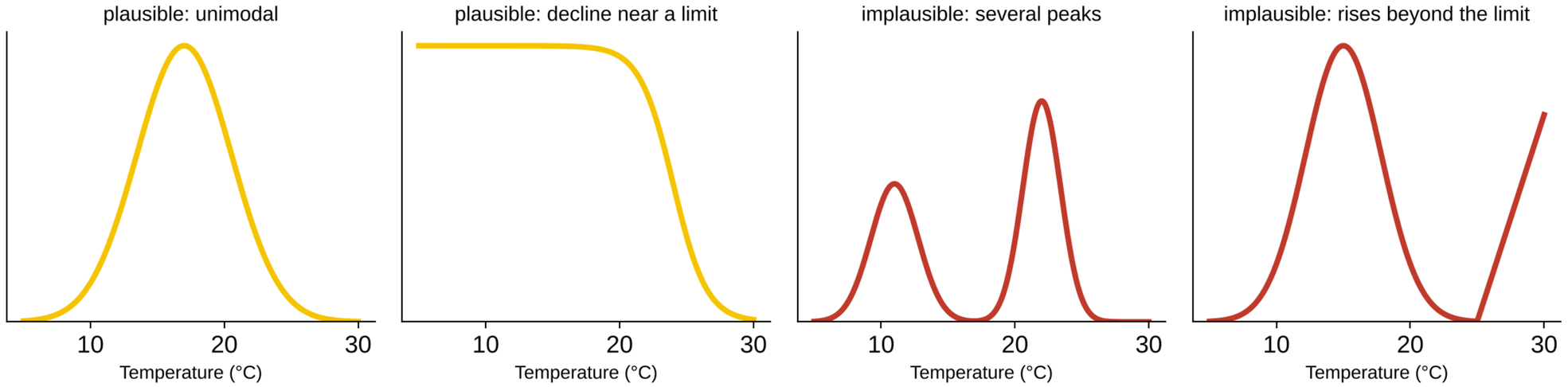


Does the model fit the expectations of ecological theory?

Partial dependency plots

Show the relationship between the probability of occurrence and each variable.

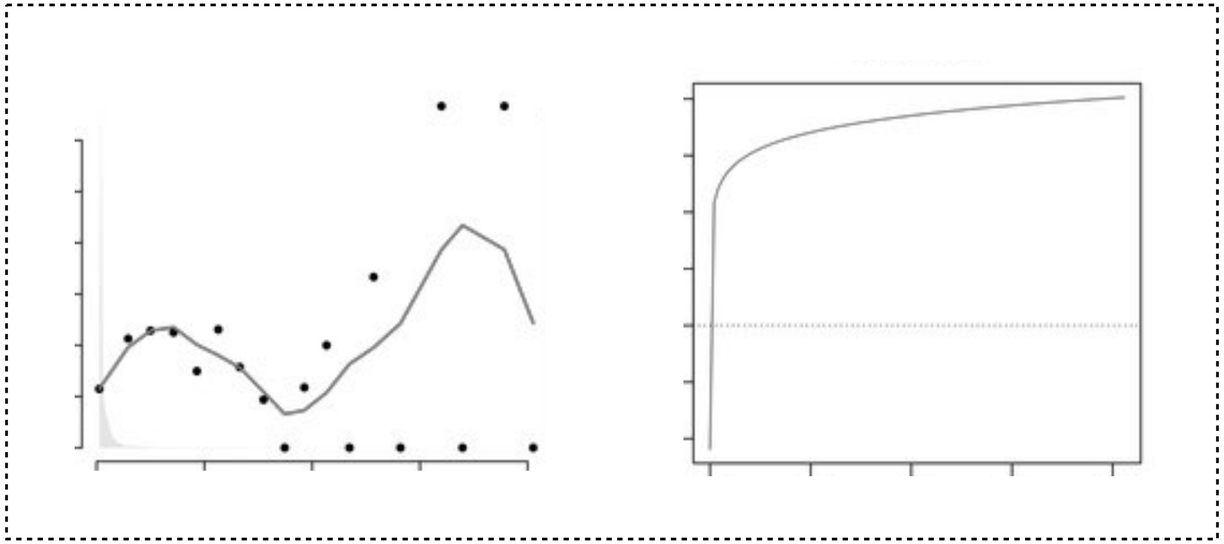
Approach: the **response is predicted across the values of one environmental variable, while the other environmental variables are held constant at their mean**.
The y-axis gives the impact on the response variable - the probability of occurrence.



Does the model fit the expectations of ecological theory?

Curve match physiological data or expectation (reliable model!)

Overfitted response vs. positive monotonicity



Forcing Ecological Rules (Monotonic Constraints)

Force the modeled relationship to follow a pre-defined, simple trend.

Prevents the model from fitting overly complex, potentially unrealistic shapes to the data for a particular variable.

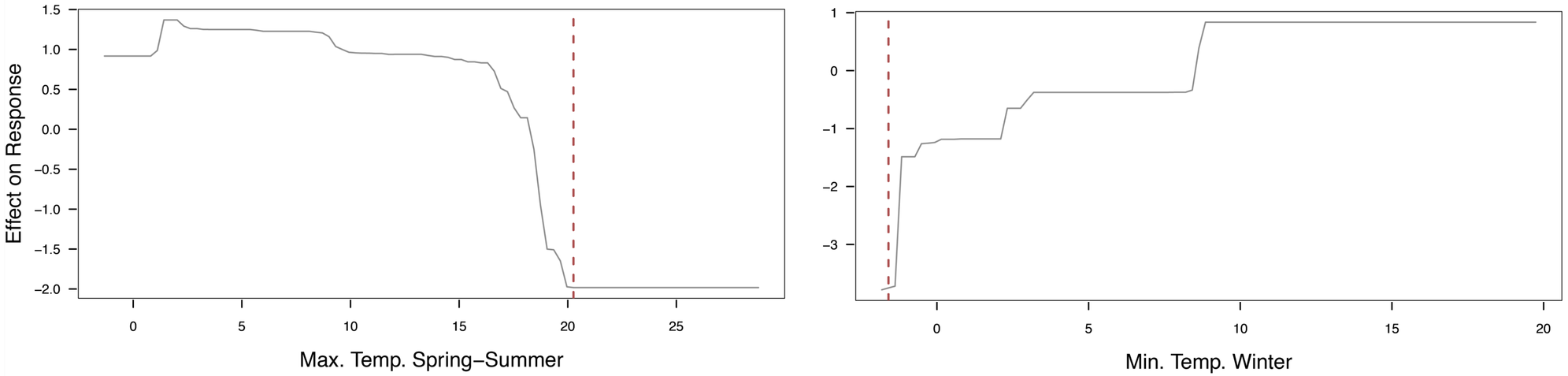
The choice of monotonicity constraints is made *a priori* per variable **based on ecological theory** (no monotonicity, negative or positive monotonicity).

Only a few machine learning algorithms have such implementation (BRT; XGBoost).

Tipping points

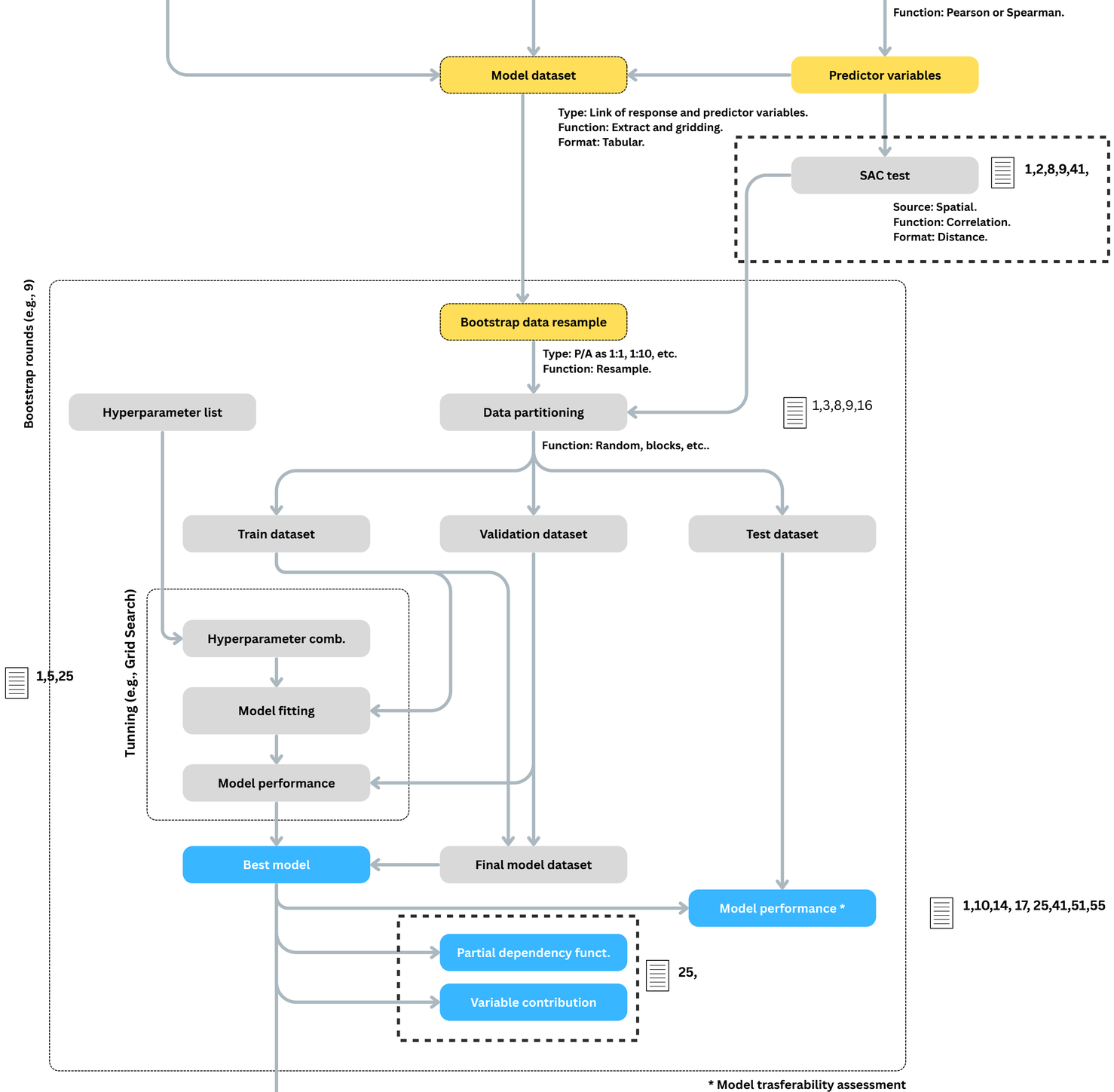
The value of a predictor at which the model shifts from suitable to unsuitable.

Extracted along each response function, using the reclassified map.



Does the model fit the expectations of ecological theory?

Limiting points match physiological data (reliable model!)


Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..
Train dataset
Validation dataset
Test dataset
Tunning (e.g., Grid Search)
Hyperparameter comb.
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
Variable contribution
25,
* Model trasferability assessment